

Response to the Associate Editor and Reviewers

Adaptive Arbitration in Command Space for Target Following under Passability Mismatch

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We thank the Associate Editor and the reviewers for their careful assessment. This revision strengthens the experimental evidence, clarifies the evaluation protocol and scope, and refines the interpretation of the results while retaining the proposed command-space arbitration mechanism. We address all concerns, questions, and requested changes below, grouping related comments where useful. Purely positive or descriptive remarks requiring no action are omitted. Quoted comments retain their original wording; identifiers refer to their positions in the complete reviews rather than the order of presentation here. References within the comments refer to the original submission; locations in our responses refer to the revised manuscript. Blue text marks substantive manuscript changes.

Associate Editor

Comment AE.1.

Three expert reviewers evaluated the manuscript and provided detailed comments and suggestions that should be carefully addressed.

R1: The reviewer finds the paper interesting and appreciates the extensive ablations and comparisons. The main concerns are the fairness of the monolithic PPO comparison, possible overfitting due to training and evaluation in the same environment, limited motivation for the hexapod embodiment, and insufficient positioning relative to prior work. The reviewer also requests clarification of several mathematical and methodological aspects.

R2: The reviewer appreciates the clear structure and thorough experiments but raises several technical and reproducibility concerns. These mainly involve the affordance map and robot footprint, the per-command risk formulation, the lateral-only Avoid expert, training details, train/test separation, and the limitations of the safety argument.

R10: The reviewer is positive about the proposed method and its experimental validation. The main concerns are the restriction to lateral avoidance, applicability beyond the structured environments considered in the paper, and missing network architecture details needed for reproducibility.

In addition to the reviewers' comments, the AE asks the authors to more clearly highlight the main contributions and novelty of the work, particularly with respect to the existing literature and closely related approaches.

Response. We clarified the contribution as learning how to allocate authority between fixed Follow and Avoid velocity proposals according to context. The Introduction now separates the problem formulation, arbitration design, and validation. Comparisons among arbitration variants reuse the same experts, and an inference ablation reuses the same Adaptive checkpoint to assess arbitration while holding expert and locomotion capability fixed. We also extended Mono training, report its training budget alongside performance, and added Fixed-Authority with its authority selected on separate validation trials. The revised manuscript makes the held-out evaluation protocol and its separation from training explicit. Related Work now distinguishes our arbitration of velocity proposals from RMP composition and MELA's composition of expert network parameters. Together, these revisions make the central contribution explicit: learning how to coordinate available behaviors to maintain both target following and obstacle clearance.

Location of changes: Abstract and Introduction; Sec. II; Secs. V-A/B and VI-A-E; Tables II and IV; Secs. VII–VIII.

Reviewer 1

Major concerns

Comment R1.1: Monolithic PPO comparison.

The ablation comparing the proposed method to a monolithic PPO baseline is unconvincing. This comparison should arguably be the central one in the paper, and the failure to present it convincingly is a significant shortcoming. The authors state that the PPO network was trained with "the same training budget as PCR-Net," without specifying what this means in practice. If the intention is to argue that the proposed method is considerably cheaper to train, this is a reasonable claim, but it should be substantiated by training the PPO baseline to its fullest potential and then presenting training cost as a separate performance metric alongside the others. As currently presented, the comparison does not achieve this.

Response. Following your suggestion, we extended Mono training from 24.576M interactions to 49.152M and 61.440M and evaluated checkpoints from all three budgets at 0.35, 0.50, and 0.60 m/s. Table IV reports the training interactions alongside task success; Fig. 5 shows how task capability emerges using probes from frozen checkpoints.

Mono shares the base sensor inputs, map encoder family, command limits, post-processing, and frozen locomotion. Mono and the gate have 1,029,447 and 1,063,237 trainable parameters, respectively, including training-only critics; Adaptive additionally uses the frozen Avoid expert. Adaptive’s 24.576M budget includes 12.288M interactions for Avoid and 12.288M for the gate, with shared locomotion training excluded. Each Mono checkpoint is held fixed across speeds and evaluated with three evaluation seeds from one training run.

Extended training achieves 94.5% task success at 0.50 m/s, showing that Mono can learn the complete task. At 0.35 m/s, success falls from 57.3% to zero between the two extended checkpoints. At 49.152M interactions and 0.60 m/s, 60.4% of episodes complete all obstacle rows without collision, yet task success remains zero with 1.25 m follow-distance MAE. This separates the ability to traverse obstacles from sustained target following in this training run. Adaptive exceeds 95% success at all three speeds at 24.576M interactions. The extended comparison therefore shows that Mono can acquire task competence, while Adaptive maintains more consistent complete-task performance across the tested speeds.

Location of changes: Sec. IV-F; Sec. VI-D; Fig. 5; Table IV.

Comment R1.2.

The simulation-based ablation is confounded by the fact that the network was trained on the same environment used for evaluation. This gives learned methods an inherent advantage, as they benefit from fine-tuning on the specific evaluation environment. This seems to be particularly visible in the accompanying video, where the learned policy seems overfit to the distance between rows of obstacles. I recommend repeating the evaluation on environments not seen during training, ideally on multiple such environments, as is standard practice in the field.

Response. In the original manuscript, “fixed hardest stage” referred to the Stage-4 evaluation setting and nominal passage pattern. Obstacle positions were perturbed around that pattern in each episode. The formal comparisons reported in the original submission used this held-out layout family, excluded from Avoid pretraining, gate training, checkpoint selection, and parameter tuning. We have revised Sec. V-A and Fig. 4 to clarify the protocol and distinguish the training and evaluation geometries.

Training uses mirrored templates with alternating left/right passages. The evaluation preserves the passability-mismatch task and introduces successive passages on the same side, breaking the regular alternation used in

training. The nominal row spacings extend beyond the range used in training. The nominal layout contains 15 obstacles (three per row) and has a minimum passage width of 0.75 m, compared with the robot’s approximately 0.65 m leg-inclusive envelope. Fig. 4(a) shows a representative training scene, whereas Fig. 4(b) shows representative trajectories with obstacles at the nominal geometry. All three target speeds use the held-out evaluation protocol: 0.35 and 0.50 m/s are within the training speed range, while 0.60 m/s additionally tests speed extrapolation.

Location of changes: Sec. V-A; Fig. 4; Table II.

Comment R1.3.

The hexapod embodiment, while featured in the title, is not well motivated in the paper. An ablation (in simulation) examining the impact of embodiment on system performance would strengthen the paper considerably—for instance, would such a complex architecture be necessary for a simpler embodiment?

Response. We revised the title and Introduction to center the contribution on arbitration in command space and clarify the hexapod’s role as the test platform. Its lateral maneuverability supports the proposed decomposition, while its body, legs, and locomotion dynamics create a concrete passability mismatch with a walking person. The gate operates on high-level velocity proposals without hexapod-specific joint observations. The revised paper therefore treats the hexapod as the validation platform for passability mismatch rather than as a requirement of the arbitration formulation.

Location of changes: Title; Sec. I.

Comment R1.4.

The related work section does not clearly establish the paper’s contribution relative to prior work. In many instances, the contrast with existing approaches is presented, but the motivation for why the proposed approach should be superior is not adequately articulated.

Response. The revised Introduction motivates the design around a simple question: how should fixed Follow and Avoid proposals share authority as their demands change? Related Work now distinguishes our arbitration of velocity proposals from target navigation, RMP composition, and MELA’s composition of expert networks. Comparisons that reuse the same experts, together with the inference ablation on the same checkpoint, test this design directly against simpler alternatives.

Location of changes: Secs. I–II; Secs. V-B and VI-B/C; Sec. VII.

Comment R1.5.

The overall system appears overengineered, and even after reviewing the ablations, I remain unconvinced that this level of complexity is necessary.

Response. Your concern about system complexity prompted us to test whether simpler coordination rules suffice. We therefore added Fixed-Authority alongside Additive-Fusion. Its global authority was chosen from 21 values between 0 and 1 on validation trials at 0.35 and 0.50 m/s, maximizing mean task success, and was then fixed for formal evaluation. These comparisons use the same analytic Follow expert, frozen Avoid expert, and locomotion policy. At 0.60 m/s, Adaptive achieves 95.3% success, compared with 66.9% for Additive-Fusion and 0% for Fixed-Authority. Disabling the learned correction in the same Adaptive checkpoint also reduces success to zero. Together, these comparisons show the value of changing authority with context under high conflict.

Location of changes: Sec. V-B; Sec. VI-A–C; Tables II–III; Sec. VII.

Minor concerns

Comments R1.6–R1.7: Writing and method naming.

- R1.6.** The writing is at times overly elaborate, which detracts from clarity and precision. Examples include the phrases "PCR-Net uses command-direction-conditioned clearance," "matching their bounded semantics while allowing state-dependent concentration," and "command risk is used as a clearance prior rather than a reach-avoid value."
- R1.7.** The choice of the name "PCR-Net" is questionable given the prior, influential work of Sarode et al. [1]; the authors should either change the name or provide explicit justification for the overlap. Furthermore, the usage of the name is inconsistent throughout the paper—at times referring to the learned gate, and at other times to the entire system. - The latter usage is particularly confusing, as the "-Net" suffix conventionally implies a monolithic network. In the Experiments section, the method is referred to inconsistently as both "PCR-Net" and "Learned-w," which adds further confusion.

Response. We shortened elaborate descriptions and removed PCR-Net as a method name, avoiding the overlap with Sarode et al. The text uses the proposed method or arbitration framework; figures and tables use Adaptive. The arbitration gate names only the learned component, and Learned-w is no longer a name for the complete method. Risk queries, gate outputs, and command blending are described directly, with consistent terminology in Fig. 2.

Location of changes: Title and Secs. I–IV; Fig. 2; figures and tables throughout.

Comments R1.8–R1.11: Conflict, map terminology, and notation.

- R1.8.** The sentence "PCR-Net addresses this conflict directly in the robot's command space. . ." does not clearly specify which conflict is being referenced. Relatedly, the statement "Standard local navigation formulations that treat the target only as a moving waypoint do not explicitly preserve target tracking during local avoidance, and fixed-priority switching between a follow and an avoid behavior can lose the target during the maneuver" describes a limitation that does not appear to be guaranteed to be resolved by the proposed solution either.
- R1.9.** The term "affordance map" used to describe the occupancy map is confusing and may benefit from reconsideration.
- R1.10.** The statement "and a single command generally cannot maximize both target recovery and immediate clearance" appears mathematically inconsistent if the two objectives are defined as orthogonal vectors. If they occupy decoupled subspaces, both could in principle be optimized independently, and therefore jointly maximized, simply through addition.
- R1.11.** The mathematical exposition is at times difficult to follow. Some variables are introduced but never subsequently used (e.g., m_t), while others are used without being introduced at all. In addition, the parameter y is used to denote both the common coordinate y and a scalar gating value, which is a source of confusion.

Response. The revised Introduction defines the conflict as the changing trade-off between target retention and collision avoidance. Sec. III-C explains that orthogonality separates command coordinates, while executed effects remain coupled through robot footprint, obstacle geometry, locomotion, command limits, and visibility. Additive-Fusion directly tests the alternative of adding the proposals. This task-specific formulation replaces the broad statement about waypoint-based navigation.

We replaced affordance map with local occupancy-free-space map, distinguishing its dimensionless free-space score from the directional distance query. The scalar authority is now α , avoiding confusion with spatial y . We

define ρ_F , ρ_A , w , and memory m_t where introduced, specify memory as a gate input, and clarify that signed modulation applies outside the deadband.

Location of changes: Secs. I–III; Secs. IV-C/D; Sec. VI-C.

Comment R1.12.

Several aspects of the method are described only briefly, leaving a number of important questions unanswered, including: Why are the gate parameters chosen in the manner described? Is there a benefit to supplying both the risk value and its memory? How exactly is the memory supplied to the gate? Why are slew-rate limiting and clearance-based scaling necessary?

Response. Secs. IV-C–F now explain the roles of the authority limits, risk memory, and command processing. The gains $\lambda = 0.30$ and $\gamma = 0.15$, together with the deadband threshold 0.05, were set during development and retained as fixed hyperparameters in training and evaluation. The gains allow a learned correction of up to 0.30 and an analytic correction of up to 0.15 before clipping, giving the learned term twice the adjustment range; the deadband suppresses corrections when $|2w - 1| \leq 0.05$. Instantaneous risk reflects the current map, while m_t retains recent hazard information and decays with forward displacement. Supplying both lets the gate distinguish current proximity from recent exposure. The memory value is one entry in the gate’s 18-dimensional goal/conflict vector before fusion with map and state features. Slew-rate limiting bounds changes between successive commands, and clearance-based scaling reduces translation near obstacles. Both operations are shared across the compared methods.

Location of changes: Sec. IV-C; Secs. IV-D–F; Sec. V-A.

Comments R1.13–R1.14: Positioning relative to related work.

R1.13. The contrast drawn with the work of Scheidemann et al. [2] is unclear. That work also addresses command-space arbitration, albeit in a less explicit manner, and this distinction should be clarified. References [3] and [4] are cited together, but only one of the two is compared against in the paper. The authors should clarify why both are not compared, or justify the omission.

R1.14. The statement "Published leader-following and agile-navigation stacks often couple different sensors, planners, and low-level controllers" would benefit from a supporting citation, as it is currently unclear what specific works are being referenced.

Response. We now distinguish the methods by what they combine. Scheidemann et al. address the closely related problem of coordinating following and obstacle avoidance through RMP accelerations and metrics, whereas our gate allocates scalar authority between fixed velocity proposals. He et al. address reach–avoid locomotion toward a static goal, and Miki et al. address perceptive terrain locomotion. He et al. and Miki et al. operate at the reach–avoid or locomotion level rather than as replacements for the high-level authority allocator, so we discuss them as related work and use alternatives with the same command interface for empirical comparison. We also removed the broad statement about published stacks.

Location of changes: Sec. II; Sec. V-B.

Comment R1.15.

The bottom row of images in Figure 6 is too small, and the accompanying text is illegible.

Response. We revised Fig. 7(b) to present three representative onboard frames side by side across the full page width, improving the visibility of the target detections and local maps. The onboard images referred to as Fig. 6 in the review now appear in Fig. 7(b).

Location of changes: Fig. 7.

Questions for the authors

Comment R1.16.

Why is a 3-DoF avoidance policy trained only for two of the output dimensions to subsequently be dropped? Would it not be more direct to train a 1-dimensional output from the outset?

Response. A one-dimensional head would indeed match the executed Avoid action more directly. In the current implementation, the pretrained Avoid expert retains a parameterization with three command outputs, while arbitration uses only its lateral mean as u_A . This parameterization is therefore not required by the arbitration formulation. We keep this pretrained expert fixed across all arbitration comparisons so that the evaluated differences arise from authority allocation rather than changes in avoidance capability.

Location of changes: Sec. IV-B; Sec. IV-F.

Comment R1.17.

In the PPO training procedure, is the map provided as an unmodified input? If so, this would deviate from standard practice for state-of-the-art PPO networks applied to this type of task, and could substantially degrade training performance.

Response. The map is processed by a CNN before fusion. Sec. IV-F specifies the two map channels plus two coordinate channels, convolution dimensions, kernels, strides, pooling, scalar encoders, and fused features for Avoid and the gate. Mono uses the same spatial CNN encoder family, with map weights learned as part of its complete policy. The same section specifies the separate training-only critics.

Location of changes: Sec. IV-F; Sec. VI-D.

Reviewer-supplied references

- [1] Sarode, Vinit, et al. "Pcnet: Point cloud registration network using pointnet encoding." arXiv preprint arXiv:1908.07906 (2019).
- [2] C. Scheidemann et al., "Obstacle-avoidant leader following with a quadruped robot," in 2025 IEEE International Conference on Robotics and Automation (ICRA), 2025, pp. 1407–1413.
- [3] T. He, C. Zhang, W. Xiao, G. He, C. Liu, and G. Shi, "Agile but safe: Learning collision-free high-speed legged locomotion," in Robotics: Science and Systems (RSS), 2024.
- [4] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning robust perceptive locomotion for quadrupedal robots in the wild," *Science Robotics*, vol. 7, no. 62, Art. no. eabk2822, 2022.

Reviewer 2

Comments R2.3–R2.5: Organization and writing.

R2.3. Introduction makes heavy use of em-dashes, sometimes it is not clear why. Please revise, fix some overly complicated sentences

R2.4. Generally, level of detail is not very consistent, consider re-structuring starting with the very fundamental ideas to the implementation details.

R2.5. Avoid inconsistent capitalization.

Response. We reduced em-dashes and shortened complicated sentences. The revised presentation moves from command decomposition to risk, arbitration, execution, and training; Table I collects architecture details. Method labels and component names now use consistent capitalization.

Location of changes: Secs. I–IV; Table I.

Comments R2.7–R2.9: Figures and supplementary video.

R2.7. The supplementary video helps in understanding the setup. Please consider adding an indicator of how much of it is sped up.

R2.8. Fig. 2 is low-res, consider vector graphics.

R2.9. Sec. IV: I do not find PCR, Gate, conflict-aware fusion etc. in Fig. 2. Consider unifying the notation between text and figure so it is easier to see which block is being discussed.

Response. We replaced Fig. 2 with revised PDF artwork and aligned its component names and α , w , and α_{eff} notation with the text. The supplementary video includes playback-speed indicators.

Location of changes: Fig. 2; supplementary video.

Comments R2.12 and R2.14: Interpretability and command bounds.

R2.12. Learning only the gating: it keeps the underlying controllers in charge and keeps the behaviour explainable., highlight explainability / deterministic more

R2.14. Bounded-residual / convex-hull argument for the fused command is a nice property to have, even tough not clear to me how this helps in terms of safety other then preventing anyway capped action magnitude

Response. The revision makes the decision structure explicit: Follow is analytic, Avoid and locomotion are frozen, and evaluation uses the gate’s Beta means, giving a deterministic output for a given input. The authority and residual show how the proposals are combined. The residual is collinear with the proposal difference, vanishes when the proposals agree, and has bounded magnitude. This constrains the correction relative to the expert proposals rather than only bounding its absolute magnitude. These are properties of the raw command blend; collision and task outcomes are evaluated empirically in simulation and hardware.

Location of changes: Sec. IV; Sec. VI; Fig. 6.

Comments R2.15–R2.17: Map construction, coverage, and side-stepping.

R2.15. The affordance map is unclear: what coverage does it have around the robot, does it see sideways, and is it a built map or instantaneous from depth? Please state this explicitly. if its a robo-centric map, state which method was used

R2.16. Related: why is side-stepping possible at all if there is no affordance information to the side? How is perception handled there?

R2.17. Is YOLO used only for the person/target, or does it also contribute semantic obstacle detection? The masking of target-associated depth points is mentioned, but a single explicit sentence on the division of labour between detection and stereo depth would help.

Response. We now specify how the map is built and why it supports sidestepping. The robot-fixed local occupancy-free-space map is represented as a 32×32 grid covering a forward $3 \text{ m} \times 3 \text{ m}$ region, with its origin at the camera mount. YOLOv8 is used only for target detection and relative goal estimation. Obstacle geometry comes from stereo depth after masking depth points associated with the detected target. The remaining valid depth points are back-projected using camera intrinsics, transformed into robot-aligned coordinates, filtered by height, and rasterized into the occupancy channel of the local map. The map is updated from current and short-term depth observations, while m_t separately stores scalar risk history; no persistent global obstacle map is maintained. Adaptive and Rule-Override use identical target detection, local map generation, and perception settings.

The forward depth view captures obstacle geometry on the left and right portions of the scene ahead of the robot, so a feasible bypass can be observed before the robot reaches the obstruction and a sidestep can be initiated in advance. This does not imply complete side-facing coverage: regions directly beside the robot may lie outside the current depth view. A 3 m query return therefore means that no occupied cell was observed in the queried cone, rather than that the full lateral path has been verified clear. The scalar risk memory m_t retains recent hazard information but does not reconstruct obstacle geometry outside the observed map.

Location of changes: Secs. IV-B/C; Sec. VI-E.

Comments R2.18 and R2.21–R2.25: Map representation, robot extent, and directional risk.

- R2.18.** No footprint / body-size awareness is visible in the risk query. Given the leg-inclusive envelope of $\sim 0.65 \text{ m}$, how is the robot extent accounted for? - in particular during rotate-on-spot command
- R2.21.** Map cell size is given as 0.09375 m ; this sub-millimetre precision is not meaningful, consider rounding or stating map extent and resolution instead.
- R2.22.** Could you elaborate on what exactly the "clearance proxy" channel is and how it is computed?
- R2.23.** Why only a 25 deg half-angle cone? If the robot is already close to an obstacle, does it not need to look further sideways for the full robot extent to be covered by the swept motion?
- R2.24.** The numeric values for free/safe ($d_{\text{safe}} = 0.27 \text{ m}$, $d_{\text{free}} = 0.57 \text{ m}$) appear without justification. Is this also where the clearance margin used in the results comes from? There is no sensitivity analysis on these values. Please consider whether they belong in the method at all, or whether they are hyperparameters to be tuned per robot/environment.
- R2.25.** The speed-scaled risk variant is presented and then discarded inside the method section (ablation-style). It is not clear which variant is used for the reported results.

Response. The dilation radius was selected from physical measurements of the hexapod, accounting for its body dimensions and the additional lateral extent introduced by leg motion during locomotion. The resulting free-space channel therefore provides the learned policies with a spatial representation that reflects the robot’s effective locomotion envelope.

We report map coverage by extent and resolution: a 32×32 grid over a forward $3 \text{ m} \times 3 \text{ m}$ region. The dimensionless free-space channel is formed by dilating occupancy, deriving a free mask, taking its cellwise maximum with a 3×3 local average, and masking occupied and unobserved cells. In simulation, occupancy dilation uses a 7×7 square kernel. Obstacle dilation thus shapes the spatial representation supplied to the learned policies, whereas the analytic directional-risk query separately uses the occupancy channel to measure metric obstacle proximity.

Robot extent is also reflected in collision checks during training and evaluation and in the envelope-intrusion criterion used for Avoid pretraining. For a nonzero translational proposal, the analytic query measures distance

from the map origin to the nearest occupied cell within a cone centered on the proposal direction. Its 25° half-angle is a fixed heuristic defining the query neighborhood. In simulation, near-zero translation, including rotation on the spot, uses the distance to the nearest occupied cell in the local map as a proximity signal. Neither the fixed dilation nor this query computes the volume swept by the body and legs during rotation.

The nominal thresholds $d_{\text{safe}} = 0.27$ m and $d_{\text{free}} = 0.57$ m are fixed design hyperparameters of the subsequent mapping from distance to risk, distinct from the dilation parameter used to construct the free-space channel. They are not derived from passage width and do not define a certified clearance margin. To assess local dependence around this nominal setting, the test summarized in Sec. V-A used $(d_{\text{safe}}, d_{\text{free}})$ pairs (0.27, 0.57), (0.22, 0.52), (0.32, 0.62), (0.27, 0.47), and (0.27, 0.67) m at all three speeds (one evaluation seed, 101; 128 episodes per setting and speed). Across mixed difficulty levels, the pooled success and collision rates varied by 1.56 and 1.30 percentage points, respectively, with the policy and post-processing fixed. This sensitivity test is separate from the formal held-out evaluation.

All reported comparisons use this fixed-threshold risk formulation without speed scaling or a braking-distance model.

Location of changes: Secs. IV-B/C; Sec. V-A; Sec. VII.

Comments R2.19–R2.20: Concavities and dynamic obstacles.

R2.19. Concave obstacles: what happens when the avoidance direction points backwards or into the concavity?

R2.20. Dynamic obstacles are not treated at all (all obstacles are static, only the target moves). At least discuss this as a limitation, ideally show one experiment / clearly mark as limitation

Response. The Discussion now states that the method assumes a feasible lateral bypass is visible ahead. Concavities, dead ends, and transverse barriers may require stopping, reversing, or higher-level replanning beyond local sidestepping. The evaluation uses static obstacles; interaction with dynamic obstacles is outside the evaluated scope.

Location of changes: Sec. IV-B; Sec. VII.

Comment R2.26.

There is no good motivation why the Avoid expert may only command lateral motion. Is backward motion not necessary for some obstacle configurations (dead ends, concave geometry)?

Response. Sec. III-B now explains the decomposition directly: Follow controls forward motion and yaw to keep the robot oriented toward the target, while Avoid supplies lateral motion around an observable bypass. Sec. VII identifies dead ends, concave traps, and a target stopped before a barrier as cases that can require stopping, backward motion, or higher-level replanning beyond this local action model.

Location of changes: Sec. III-B; Sec. IV-B; Sec. VII.

Comments R2.27–R2.31: Training, observations, and network architecture.

R2.27. It is not clear how the Avoid expert is trained. Please state the reward function, observations and termination conditions.

R2.28. When the Avoid expert is trained, is the Follow expert already in the loop? What kind of tuning happens in that stage?

R2.29. which simulator setup is used?

R2.30. The simulator and learning stack are mentioned only for the gating (Isaac Gym, RSL-RL, PPO) please state clearly which simulator/asset setup is used for each of the three training stages.

R2.31. A table listing, for each learned module (Avoid expert and Gate), the exact observations and actions would help a lot. Does the gate observe only the risks, or more? What exactly is "robot state" in the gate observation? Stating this clearly would considerably improve reproducibility.

Response. All three learned stages use the same hexapod asset in Isaac Gym. Locomotion is pretrained with RSL-RL and frozen; Avoid and the gate use PPO. Avoid is trained independently in randomized obstacle layouts. Passage difficulty varies by stage, the environment supplies forward motion, and Follow is absent. PPO updates Avoid while locomotion remains fixed.

Gate training uses four curriculum levels: two rows at 0.25–0.40 m/s, two rows at 0.35–0.50 m/s, three rows at 0.30–0.50 m/s, and four to five rows at 0.35–0.50 m/s. Target speed is fixed within each episode; over 120,000 episodes, sampling shifts toward harder levels while retaining easier episodes.

Sec. IV-B gives the reward function, coefficients, activation conditions, and termination rules. The dense reward combines progress, improved directional clearance, penalties on the magnitude and variation of the lateral command, and a weak directional preference. Terminal rewards replace it: +2 for crossing the layout exit without failure and –20 for collision, envelope intrusion, falling, or tilt/height violation, with failure taking precedence. Timeout alone has no failure penalty.

Table I and Sec. IV-F give the CNN, scalar encoders, fusion, heads, critics, and budgets. Avoid receives a map, 14-D state, and 2-D goal; the gate receives a map, 13-D state, and 18-D goal/conflict vector, including commands, risks, clearances, and memory. The text enumerates the fields, specifying body-frame velocity, previous base authority α , and the formula for the conflict score. Avoid’s authority slot is zero. Its three-dimensional command distribution is evaluated through the lateral mean; four scalar gate heads form two Beta distributions for α, w , evaluated through their means. Information on row release supplies auxiliary labels, not actor observations.

Location of changes: Sec. IV-B; Sec. IV-F; Table I.

Comment R2.32.

The convex-hull / bounded-residual guarantee is nice, but: just because each individual expert command is reasonable, does that imply the mixture is safe? This feels like a strong assumption, i.e. that nudging the command slightly is enough to make it safe-ish. Please discuss the limits of this argument, in particular in the high-conflict cases where the gate is most active.

Response. Sec. IV-G now states the guarantee at the command level. The raw blend stays within the proposals’ convex hull, and the residual remains bounded and collinear with the proposal difference, including during high conflict. The set of safe executed motions need not be convex, so these properties of the commands do not certify collision avoidance.

Location of changes: Sec. IV-G; Sec. VII.

Comment R2.33.

"Fixed hardest stage" of the curriculum: are these environments that were seen during training, or held-out layouts? Please make the train/test separation explicit.

Response. The formal evaluation uses the held-out Stage-4 layout family described in our response to R1.2. The family is excluded from Avoid pretraining, gate training, checkpoint selection, and parameter tuning. Each instance is generated by randomly perturbing obstacle positions around the same nominal structure. Sec. V-A specifies its passage arrangement; Table II and Fig. 4(b) use this evaluation protocol, while Fig. 4(a) illustrates the training distribution. Reported variability is measured across three evaluation seeds under the same protocol.

Location of changes: Sec. V-A; Fig. 4; Table II.

Comments R2.34–R2.36: Failure modes, target speed, and Mono evaluation.

- R2.34.** Please discuss failure modes more / when and why does it fail, which limitation triggers failure, does it recover
- R2.35.** Only the gate is velocity-dependent, correct? It is unclear which component is expected to break when exceeding the training speed, since the gating behaviour probably has no strong velocity gradient during training. Because this above-training speed carries much of the evaluation argument, please elaborate.
- R2.36.** Monolithic PPO is only evaluated at 0.35 m/s over increasing difficulty, not across target speeds. Evaluating it at 0.50/0.60 m/s like the other methods would make the comparison more complete and would be an interesting addition. - nice to have, not crit

Response. We expanded Sec. VI to analyze how the arbitration variants fail rather than only reporting their aggregate scores. The comparisons show that lower collision, substantial obstacle progress, or low follow-distance error can each coexist with task failure. The inference ablation tests the effect of disabling the learned correction in the same checkpoint: at 0.60 m/s, setting $\Delta\alpha_w = 0$ preserves substantial row progress but eliminates task success and increases follow-distance error.

We also extended Mono evaluation to all three speeds and to matched and extended training budgets. Extended training reaches 94.5% task success at 0.50 m/s, confirming that the complete task is learnable without expert decomposition. At 49.152M interactions and 0.60 m/s, 60.4% of episodes complete all obstacle rows without collision, yet task success remains zero with 1.25 m follow-distance MAE, separating obstacle traversal from sustained following in this training run.

The 0.60 m/s condition stresses the complete following–avoidance system rather than the gate alone: Avoid and the gate observe body-frame planar velocity, while the analytic Follow proposal changes with the relative target state. On hardware, Adaptive’s three failures comprise two contacts and one target loss. Either event makes the trial unsuccessful; subsequent recovery does not change this classification.

Location of changes: Secs. VI-A–E and VII; Tables II–IV and VI.

Comments R2.37–R2.39: Related literature and design choices.

- R2.37.** The literature review covers only geometric and control-oriented work. Human following and human-robot interaction literature is not mentioned, although the target here is a walking person.
- R2.38.** Some literature backing for the design choice of reward shaping vs. constrained RL would be nice.
- R2.39.** The mixture-of-experts framing is grounded almost exclusively in very old work (Jacobs et al. 1991). Please add more recent work on learned gating / mixture-of-experts composition. where has this been used, why is it the go-to technology

Response. We added HRI work on visibility, interpersonal distance, and socially acceptable motion, together with constrained policy optimization and a recent legged mixture-of-experts example. Reward shaping expresses the graded trade-off between following and clearance in our training objective, whereas CPO maximizes return under an explicit cost constraint. MELA composes expert network parameters. Learned gating is appropriate here because Follow and Avoid already provide usable, interpretable command proposals; learning can focus on how their authority should change with context rather than relearning either behavior.

Location of changes: Sec. II; Sec. IV-F; Secs. V-B–VI.

Reviewer 10

Required Clarifications

Comment R10.1.

Longitudinal Avoidance & Continuous Obstacles: Constraining the Avoid expert strictly to lateral motion presumes all collisions can be averted by lateral sidestepping. Please discuss how the framework behaves when encountering continuous transverse obstacles (such as solid walls, dead-ends, or when the target stops abruptly in front of a barrier) where lateral motion alone cannot resolve the collision.

Response. We clarified that lateral-only Avoid requires a feasible bypass observable ahead. Solid transverse walls, dead ends, concave traps, and a target stopping before a barrier may require stopping, backward motion, or higher-level replanning.

Location of changes: Sec. IV-B; Sec. VII.

Comment R10.2.

Environment Scope Discussion: The simulation benchmarks and hardware tests rely primarily on periodic, staggered-row geometries. While staggered rows isolate lateral passability mismatch, please explicitly discuss the operational boundaries of this formulation when navigating unstructured, randomly scattered clutter or non-convex obstacles.

Response. The held-out layout breaks the regular passage alternation used in training, and the hardware tests use boxes and cones in staggered-row and 90° corridors with the same deployed policy. These tests match the method’s operating assumption that a feasible lateral bypass is visible in the forward map. In unstructured or non-convex clutter, cases that require backward escape or global replanning exceed this local action model.

Location of changes: Secs. V-A and VI-E; Fig. 4; Sec. VII.

Comment R10.3.

Network Architecture & Reproducibility: Section IV-F mentions that encoders feed a 256–256 ELU trunk with Beta heads for y and w . Please provide the exact architectural details of the local affordance map encoder (e.g., 2D CNN kernel sizes, strides, channel depths, or MLP projection dimensions) and specify how spatial map features are concatenated with the 31 scalar state inputs before entering the ELU trunk.

Response. Following your request for architectural details, we specify the map CNN, kernels, strides, padding, pooling, and projection in Sec. IV-F. The 13-state and 18-target/conflict vectors are encoded to 64 and 32 features; these join 128 map features and one difficulty input fixed to zero to form the gate’s 225-dimensional input. The ELU trunk, four Beta-parameter heads, and separate training-only critic are specified, with parameter counts that include the critic.

Location of changes: Sec. IV-F.

Adaptive Arbitration in Command Space for Target Following under Passability Mismatch

Abstract—A legged robot following a person through clutter may encounter passages that the person can traverse but the robot cannot safely negotiate. This passability mismatch creates conflicts between target recovery and obstacle clearance. We propose an adaptive arbitration framework in command space that coordinates analytic forward-yaw following and pretrained lateral avoidance proposals. A learned arbitration gate adjusts their relative authority based on risk while keeping both experts and the locomotion policy fixed, concentrating learning on behavioral coordination. On the held-out evaluation, the method achieves 95.3% task success at 0.60 m/s, 20% above the maximum training speed. With onboard target detection and stereo depth, it succeeds in 37 of 40 hardware trials across staggered-row and sharp-turn corridors. Comparisons with additive fusion, fixed authority, and monolithic PPO support the value of context-dependent authority allocation in the tested conditions.

I. INTRODUCTION

A moving person can pass through gaps that a legged robot’s gait envelope cannot safely traverse (Fig. 1). This *passability mismatch* couples collision avoidance with target retention: tracking the person can reduce clearance, whereas avoidance can cause target loss. We study how to allocate authority between Follow and Avoid proposals as this conflict changes.

The proposed arbitration framework combines analytic Follow and learned Avoid proposals in orthogonal command subspaces (Fig. 2). The gate uses obstacle risk along each proposal’s direction to allocate authority between them. The experts and the pretrained locomotion policy remain fixed, concentrating learning on their coordination.

The hexapod supports the lateral motion required by this decomposition, while its fixed 18-joint locomotion policy realizes the commands. Arbitration uses high-level velocities without hexapod-specific joint observations. Staggered rows provide a controlled stress test: successive obstacle rows require sidestepping while the target advances.

Contributions.

- **Command-space formulation of passability mismatch.** We formulate the conflict between moving-target retention and local collision avoidance as continuous authority allocation in command space.
- **Risk-conditioned adaptive arbitration.** A risk-conditioned gate blends fixed Follow and Avoid proposals in orthogonal command subspaces. The bounded raw convex blend passes through shared post-processing and fixed locomotion.

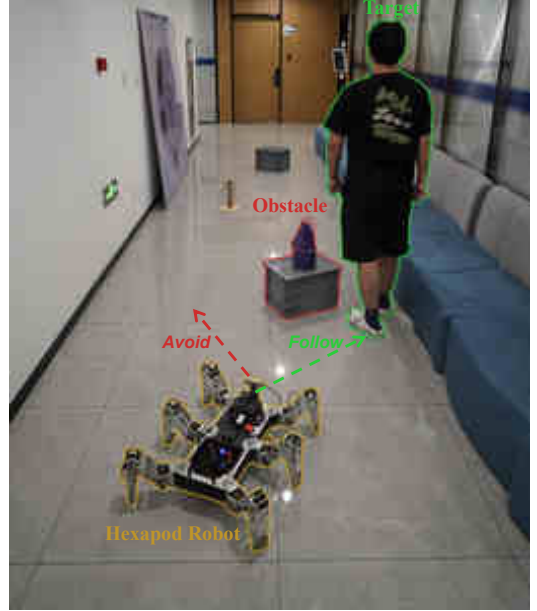


Fig. 1. Real corridor setting for hexapod target following in clutter. Gaps passable by the target may be too narrow for the robot’s leg-inclusive envelope, creating a conflict between target recovery and collision avoidance.

- **Controlled validation of arbitration.** Comparisons among arbitration variants reuse the same experts, and the inference ablation reuses the same checkpoint, keeping expert and locomotion capability fixed. Held-out simulation and hardware trials quantify the safety-following trade-off.

II. RELATED WORK

Following, avoidance, and agile locomotion. Person-following is commonly formulated as mobile-robot tracking and local navigation [1], while human-robot interaction (HRI) studies additionally consider visibility, interpersonal distance, and socially acceptable motion [2]. He et al. [3] coordinate agile and recovery behaviors through a reach-avoid formulation for collision-free locomotion toward a static goal, and Miki et al. [4] learn perceptive locomotion from exteroceptive and proprioceptive inputs. Our focus is maintaining tracking of a moving target during local obstacle avoidance.

Planning, safety, and expert arbitration. Dynamic window methods score clearance and progress over a short horizon [5]; we include a DWA-style alternative using the same command interface in Sec. VI. Control barrier functions en-

code constraints in online QPs [6]. Constrained policy optimization maximizes return under explicit cost constraints [7]; our training expresses the following-clearance trade-off through reward shaping. For behavior composition, classical mixture-of-experts methods learn gating functions [8], while MELA composes expert network parameters [9]. Scheide-mann et al. [10] combine leader-following and obstacle avoidance through RMP accelerations and metrics. Shared control methods blend policies according to context [11], while residual RL learns additive action corrections [12]. Our gate instead learns scalar authority between fixed Follow and Avoid velocity proposals. Keeping both experts fixed focuses gate training on when and how strongly to favor each behavior.

III. PROBLEM FORMULATION

A. Setting. The controlled stress test uses obstacle rows with lateral passages (Fig. 1). The high-level controller outputs a velocity command $u = [u_x, u_y, u_\omega]^\top$ (lateral, forward, yaw) at 10 Hz, realized by a fixed low-level locomotion policy at 50 Hz. The robot perceives a local occupancy-free-space map and the relative target state; it has no global map.

B. Orthogonal command decomposition. We assign following and avoidance to orthogonal command subspaces. The Follow expert acts in the forward-yaw subspace and the Avoid expert in the lateral subspace,

$$\mathcal{U}_F = \{[0, u_y, u_\omega]^\top\}, \quad \mathcal{U}_A = \{[u_x, 0, 0]^\top\}, \quad (1)$$

so that $\mathbb{R}^3 = \mathcal{U}_F \oplus \mathcal{U}_A$ and $\mathcal{U}_F \perp \mathcal{U}_A$. A Follow command $u_F = [0, u_{F,y}, u_{F,\omega}] \in \mathcal{U}_F$ is computed analytically; the learned Avoid expert produces a raw command projected onto the lateral subspace, $u_A = [u_{A,x}, 0, 0] \in \mathcal{U}_A$. Because yaw lies in the Follow subspace, a nonzero Follow authority preserves part of the yaw toward the target during lateral avoidance, helping the robot keep facing the target while it sidesteps.

C. The conflict. The proposals can be added algebraically, but their executed effects are coupled by robot footprint, obstacle geometry, locomotion dynamics, command limits, and target visibility. Tracking can therefore reduce clearance, whereas avoidance can increase target error. Arbitration regulates the relative authority of the proposals as this conflict changes.

D. Task success and diagnostics. We separate event success from trajectory diagnostics. A trial succeeds only if, at task completion, the robot has cleared the final obstacle row, lies within the follow-distance band of [1.0, 2.2] m, and has remained collision-free throughout the episode. We also report trajectory diagnostics: follow-distance MAE (mean $|\text{dist}(\text{target}, \text{robot}) - d_{\text{des}}|$ over high-level steps, with $d_{\text{des}} = 1.5$ m), L1 (no collision), L2 (target retained: follow-distance MAE ≤ 0.5 m and in-FOV ≥ 0.90), and L3 (final-row clearance). The distance condition is checked at task completion; follow-distance MAE and L2 summarize tracking over the trajectory, and L2 is not an additional success condition.

IV. METHOD

The controller combines fixed locomotion, orthogonal expert proposals, and a learned arbitration gate (Fig. 2).

A. Follow expert (analytic). The Follow expert computes $u_F = [0, u_{F,y}, u_{F,\omega}]$ analytically from the 2-D relative goal representing the target position relative to the robot, using a turn-then-advance rule with hysteresis: when the bearing magnitude exceeds 0.35 rad the forward term is suppressed and the robot turns in place; it resumes advancing once the bearing magnitude falls below 0.18 rad.

B. Avoid expert (learned). Avoid is pretrained independently in randomized obstacle layouts. Passage difficulty varies by stage, the environment supplies forward motion, and Follow is absent. The pretrained actor retains three command outputs, while arbitration uses only its lateral mean as the frozen Avoid proposal u_A (observations and architecture: Sec. IV-F, Table I).

The reward encourages forward progress and improved directional clearance, penalizes the magnitude and variation of the lateral command, and adds a weak directional preference when both lateral alternatives are sufficiently clear. At $\Delta t = 0.1$ s, the dense reward is

$$r_t^A = \Delta y + 0.03q_0(q_0 - q_u) - 0.005|u_x^{\text{raw}}| - 0.01|u_{x,t}^{\text{exec}} - u_{x,t-1}^{\text{exec}}| + 0.0005Gp\frac{u_{x,\max}^{\text{exec}}}{u_{x,\max}}, \quad (2)$$

where Δy is forward displacement and $q_j = \text{clip}((0.57 - d_j)/0.57, 0, 1)$ for $j = 0, u$. Distances d_0, d_u use cones with a 25° half-angle around the straight-ahead direction and the raw command direction on the same map before action execution. Superscripts raw/exec denote commands before/after processing; $u_{x,\max}$ is the lateral limit. The preference p is the normalized lateral component of a valid forward goal direction. The indicator $G = 1$ only beyond Avoid stage 1, with a valid preference, $\Delta y > 0$, $d_0 < 0.57$, $d_L, d_R \geq 0.57$, and $|d_L - d_R| \leq 0.10$ (distances in metres); otherwise $G = 0$. Here d_L, d_R query lateral commands $\pm u_{x,\max}$ combined with the supplied forward command. The reward does not use row indices, gap centers, or passage templates.

Terminal rewards replace r_t^A : +2 for the robot center crossing the layout's exit line without failure; -20 for collision, envelope intrusion, falling, or tilt/height violations, with failure taking precedence. Timeout ends the episode without the failure penalty. Lateral avoidance assumes that a feasible bypass is observable ahead.

C. Directional command risk. The gate receives a two-channel robot-fixed local occupancy-free-space map, represented as a 32×32 grid covering a forward 3 m $\times$ 3 m region, with its origin at the camera mount. The occupancy channel marks occupied cells. The free-space channel is a dimensionless score formed by dilating occupancy with a 7×7 square kernel in simulation, deriving a free-space mask, taking its cellwise maximum with a 3×3 local average, then masking occupied and unobserved cells; it is not a metric distance field. The dilation radius was selected from physical measurements of the hexapod to account for its body dimensions and the additional lateral extent introduced

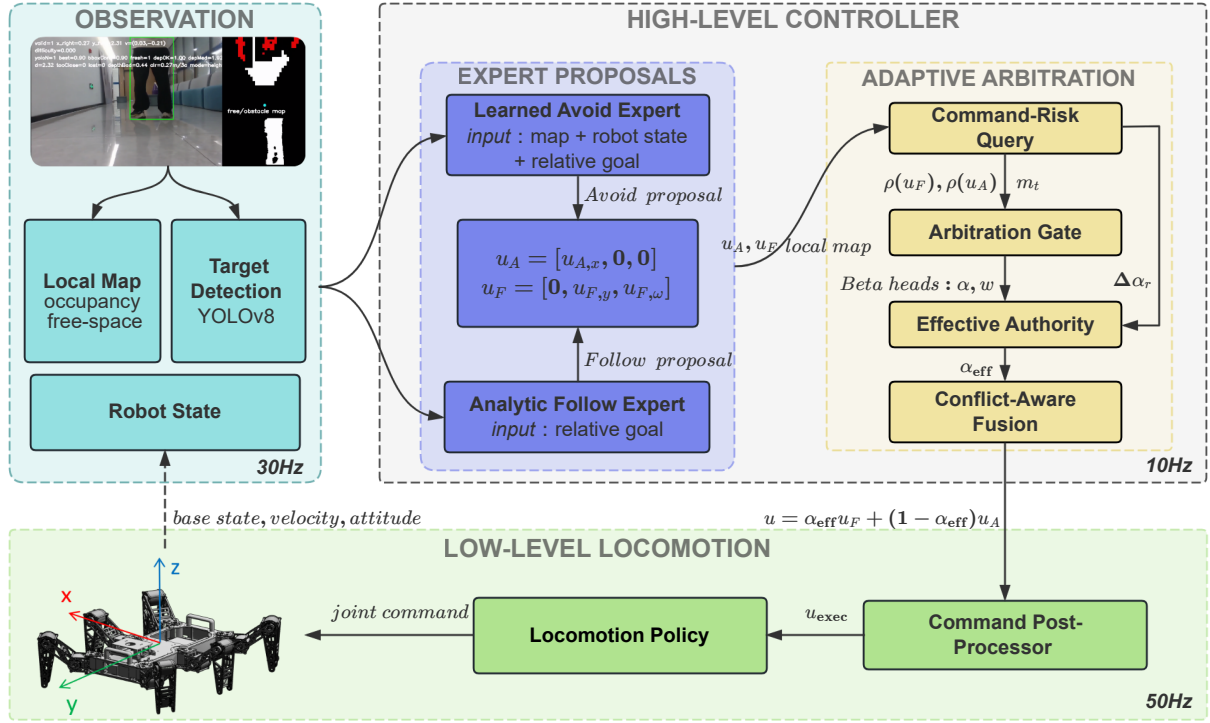


Fig. 2. Adaptive arbitration in command space. Onboard perception supplies a local **occupancy-free-space** map and relative target state (YOLOv8). The analytic Follow expert proposes forward-yaw motion, the learned Avoid expert proposes lateral motion, and directional risk **and its scalar memory** feed the arbitration gate. The gate computes α_{eff} for command fusion before shared command post-processing and a fixed locomotion policy. Rates indicate deployment frequencies: perception 30 Hz, arbitration 10 Hz, and locomotion 50 Hz.

by leg motion during locomotion. For each proposal with nonzero translation, distance d from the map origin to the nearest occupied cell in the occupancy channel is queried within a cone centered at $\text{atan2}(u_x, u_y)$ with a 25° half-angle ($d = 3.0\text{m}$ when no occupied cell is observed in the cone). When translation is near zero, the simulation query falls back to the distance to the nearest occupied cell in the local map. Neither query evaluates swept yaw motion. Sidestepping can begin using bypass space observed ahead, but an empty query cone does not establish that the entire lateral path is clear. Risk is

$$\rho(u) = 1 - \text{clip}\left(\frac{d - d_{\text{safe}}}{d_{\text{free}} - d_{\text{safe}}}, 0, 1\right), \quad (3)$$

with $d_{\text{safe}} = 0.27\text{m}$ and $d_{\text{free}} = 0.57\text{m}$. The fixed thresholds define a directional proximity score rather than clearance of the swept robot body. The 25° half-angle is a fixed heuristic that defines the directional query neighborhood; no speed scaling or braking-distance model is used.

For each proposal, $\rho_F = \rho(u_F)$ and $\rho_A = \rho(u_A)$. Risk memory with distance decay. To retain recent hazards, scalar risk memory decays with forward body displacement, with no decay during pure lateral motion or stops:

$$\begin{aligned} m_t &= \max(\rho_{F,t}, m_{t-1} e^{-\Delta s / L_{\text{clear}}}), \\ \Delta s &= \max(v_{y,t}^{\text{body}}, 0) \Delta t, \end{aligned} \quad (4)$$

with $L_{\text{clear}} = 0.40\text{m}$. This scalar is supplied to the gate as a derived conflict feature. **Instantaneous risk reflects current**

map observations; m_t retains a scalar hazard signal, not an obstacle map.

D. Gate and conflict-aware fusion. A single arbitration gate uses Beta heads for the **base Follow authority** $\alpha \in (0, 1)$ and learned conflict variable $w \in (0, 1)$. The gate combines local geometry, robot and target states, and conflict features derived from the proposals; the exact observations are specified in Sec. IV-F. The learned signed modulation and analytic risk correction are

$$\Delta\alpha_w = \lambda \text{deadband}(2w - 1), \quad \Delta\alpha_r = \gamma(\rho_A - \rho_F), \quad (5)$$

$$\alpha_{\text{eff}} = \text{clip}(\alpha + \Delta\alpha_w + \Delta\alpha_r, 0, 1), \quad (6)$$

with $\lambda = 0.30$, $\gamma = 0.15$, and $\text{deadband}(z) = z$ for $|z| > 0.05$ and 0 otherwise. The fused command is

$$u = \alpha_{\text{eff}} u_F + (1 - \alpha_{\text{eff}}) u_A. \quad (7)$$

Outside the deadband, values $w > 0.5$ reinforce following, whereas $w < 0.5$ suppresses it through $\Delta\alpha_w$. In the active unsaturated region, $\partial\alpha_{\text{eff}}/\partial w = 2\lambda = 0.60$; it is zero inside the dead zone or after clipping. **The gains bound the learned and analytic corrections by 0.30 and 0.15, respectively, before clipping.**

E. Low-level locomotion interface. The fixed locomotion policy tracks high-level commands at 50 Hz and is shared across methods. Before execution, all methods use the same amplitude clamping, slew-rate limiting **to bound command changes**, and clearance-based scaling **to attenuate translation near obstacles**.

TABLE I

NETWORK ARCHITECTURES OF THE AVOID EXPERT AND ARBITRATION GATE. BOTH USE THE SAME MAP ENCODER ARCHITECTURE WITH INDEPENDENT WEIGHTS; PARAMETER COUNTS INCLUDE THE CRITICS USED DURING TRAINING

Block	Avoid expert	Arbitration gate
Map encoder	2-ch local occupancy/free-space map + 2 normalized coordinate channels; Conv $4 \rightarrow 32 \rightarrow 64 \rightarrow 128$ (3×3 , strides $2/2/1$, pad 1, ELU); adaptive 4×4 average pooling; FC $2048 \rightarrow 128$.	
Scalar encoders	State MLP $14 \rightarrow 64 \rightarrow 64 \rightarrow 64$; goal MLP $2 \rightarrow 32 \rightarrow 32$ (ELU).	State MLP $13 \rightarrow 64 \rightarrow 64 \rightarrow 64$; goal/conflict MLP $18 \rightarrow 32 \rightarrow 32$ (ELU; 31 scalar inputs total).
Fusion	Concat. $128 + 64 + 32 + 1 = 225$; MLP $225 \rightarrow 256 \rightarrow 256$ (ELU). The 1-D difficulty input is fixed to zero for the evaluated gate.	
Actor head	Mean $256 \rightarrow 64 \rightarrow 3$; std $256 \rightarrow 64 \rightarrow 3$ (Softplus); lateral mean executed.	$4 \times (256 \rightarrow 64 \rightarrow 1)$ Softplus concentration heads; two Beta distributions for α, w ; Beta means used at evaluation.
Critic / params	Value $256 \rightarrow 64 \rightarrow 1$; 1,029,575 params.	Value $256 \rightarrow 64 \rightarrow 1$; 1,063,237 params.

F. Training procedure. We first pretrain and freeze a hexapod locomotion policy conditioned on velocity commands with RSL-RL [13], using reward terms for camera stability and smooth impacts. With Avoid also frozen (Sec. IV-B), only the arbitration gate is trained. An auxiliary binary cross-entropy loss with weight 0.05 encourages $w = 0$ in states that clearly favor avoidance: cases before row release satisfying $\rho_F > 0.25$, $\rho_F - \rho_A > 0.05$, and command cosine < 0.5 , or global cases satisfying $\rho_F > 0.45$, $\rho_F - \rho_A > 0.15$, and command cosine < 0.3 .

The Avoid expert, locomotion policy, and gate use the same hexapod asset in Isaac Gym [14]; locomotion uses RSL-RL and the two learned high-level policies use PPO [15]. The curriculum has four levels and varies the number of obstacle rows from two to five together with target speed from 0.25 to 0.50 m/s, with speed fixed within each episode. Over 120,000 episodes, sampling shifts toward harder levels while retaining easier episodes. The five-row training layouts use two mirrored templates with alternating left/right passages and independent perturbations of obstacle positions up to ± 0.06 m; Sec. V-A describes the held-out evaluation. Training obstacles are [static, with a radius of 0.15 m and a height of 0.50 m](#); contact friction is randomized in $[0.4, 0.8]$. During training, the policy receives a local occupancy-free-space map rasterized from simulated obstacle geometry and restricted to the D435i stereo FOV rather than rendered depth.

Table I summarizes the architectures; critics have separate encoders and trunks. The gate’s 13-D robot state comprises planar position in world coordinates, yaw, body-frame planar velocity, yaw rate, height, roll/pitch, previous base authority α , and the last command. Avoid fixes the authority

entry to zero and appends imposed forward speed (14-D state); its map and 2-D relative goal are separate inputs. The gate’s 18-D goal/conflict vector comprises the relative goal, Follow/Avoid proposals and their difference (9), risks ρ_F, ρ_A , risk memory m_t , translational command cosine c , conflict score $\max(\rho_F - \rho_A, 0)(1 - c)/2$, and two clearances normalized as $\text{clip}(d/(3\text{ m}), 0, 1)$. Here $c = 1$ if either translational proposal has norm at most 10^{-6} .

Both high-level stages use 1,000 iterations (12.288M interactions) each, for 24.576M in total; shared frozen locomotion training is excluded. Gate PPO uses 256 environments with 48 steps per iteration; Avoid uses 512 environments with 24 steps. The gate is optimized for five epochs per iteration with minibatches of 12,288, learning rate 6×10^{-5} , entropy coefficient 0.04, and clip ratio 0.05. Gate training combines row-progress, follow-distance, success, and collision/failure/time rewards with the w auxiliary loss. [Information on row release supplies auxiliary training labels only](#); the actor instead receives scalar risk memory.

G. Structural properties and scope. The raw blend lies in the proposals’ convex hull. With $u_{\text{nom}} = \alpha u_F + (1 - \alpha)u_A$ and $\delta u = u - u_{\text{nom}}$,

$$\begin{aligned} \delta u &= (\alpha_{\text{eff}} - \alpha)(u_F - u_A), \\ \|\delta u\| &\leq (\lambda + \gamma)\|u_F - u_A\|. \end{aligned} \quad (8)$$

Thus, the correction vanishes at coincident proposals and is bounded during conflict. [These properties concern raw commands; they do not guarantee collision avoidance, since the set of safe executed motions need not be convex.](#)

V. EXPERIMENTS

A. Setup. Formal comparisons use a held-out five-row Stage-4 layout family excluded from Avoid/gate training, checkpoint selection, and parameter tuning. It preserves passability mismatch while breaking training’s left/right alternation with successive passages on one side. The nominal row spacings extend beyond the range used in training. Each episode randomly perturbs obstacle positions around the nominal pattern. The nominal pattern has 15 obstacles (three per row) and a 0.75 m minimum passage width versus the robot’s ~ 0.65 m leg-inclusive envelope. Tests use 0.35, 0.50, and 0.60 m/s; 0.60 m/s is 20% above the 0.50 m/s training maximum. Main results are reported as mean $\pm$ SD over three evaluation seeds, with 128 episodes per seed (384 per method and speed; 50-s horizon). The hexapod has 18 actuated joints and a 0.25 m passing half-length. The D435i uses $87^\circ \times 58^\circ$ depth FOV over 0.28–3.0 m. Simulation uses structured geometric primitives; inference ablation and DWA use the same evaluation protocol. [For Mono, curriculum probes use frozen checkpoints at levels 2–4 to diagnose training capability](#); these probes are separate from the formal held-out evaluation. A separate sensitivity test across mixed difficulty levels shifted both nominal risk thresholds by ± 0.05 m or varied d_{free} alone by ± 0.10 m, with the policy and post-processing fixed. Across three speeds (128 episodes per setting and speed), the pooled success and collision rates varied by 1.6 and 1.3 percentage points, respectively.

B. Compared methods. We organize the comparison into three groups. (1) *Arbitration ablations* share the experts, base sensor inputs, and execution stack but apply different arbitration rules: *α -only* ($\alpha_{\text{eff}} = \alpha$, no risk or learned modulation); *Geom-w*, a geometric clearance heuristic $w_{\text{geom}} = e^{-d/\tau}$ using the clearance d along the Follow command and $\tau = 0.15$ m, with $\alpha_{\text{eff}} = \alpha(1 - w_{\text{geom}})$; *Risk-only*, the learned base α plus analytic $\Delta\alpha_r$ term, with $\alpha_{\text{eff}} = \text{clip}(\alpha + \Delta\alpha_r, 0, 1)$, retrained from scratch without w ; *Rule-Override*, a reactive override on $\rho_F - \rho_A$ that forces lateral avoidance by cutting the forward command to as low as 10% while retaining up to 70% of the yaw command, using one fixed parameter set across all speeds; and *Adaptive*, the full proposed method. (2) *Structural alternatives*. *Additive-Fusion* replaces the gate with direct command addition, $u = u_F + u_A$, keeping the same experts but with no gate. *Fixed-Authority* uses $u = \alpha^* u_F + (1 - \alpha^*) u_A$ with a global constant $\alpha^* = 0.50$ selected from a 21-point validation sweep over $[0, 1]$ by maximizing mean task success across separate validation trials at 0.35 and 0.50 m/s, then held fixed across speeds, with the same experts, post-processing, and locomotion policy. (3) *External alternatives* (Sec. VI): a *monolithic PPO* policy and a *DWA-style velocity search*, both using the same base map, robot state, and target sensor inputs, command limits, and fixed low-level stack.

C. Metrics. Task success is the event-level criterion of Sec. III-D; we report collision rate and progress toward final-row clearance as trajectory diagnostics, plus follow-distance MAE and L2 where target retention is relevant. Row progress is the fraction of obstacle rows safely completed, distinct from the binary final-row-clearance indicator L3.

VI. RESULTS

All simulation results use the event-level success definition of Sec. III-D; their evaluation protocols are specified in Sec. V-A.

A. Main performance. At high conflict, Adaptive combines high task success with low collision (Table II; Fig. 3). Fig. 4(a) shows a scene from the training distribution, distinct from the formal evaluation layout. At 0.60 m/s, success is 95.3%, collision is 3.9%, and follow-distance MAE is 0.62 m.

The arbitration variants degrade differently as target speed increases. Although α -only learns a context-dependent base authority, its task success falls to zero at the two higher speeds. Geom-w and Risk-only remain effective at the intermediate speed, but at 0.60 m/s their lower collision rates than α -only coexist with near-zero task success and large follow-distance errors. Rule-Override retains substantial task success but incurs frequent collisions. Across these variants, task failure appears through different combinations of collision and follow-distance error.

B. Inference ablation. Table III reuses the Adaptive checkpoint while setting $\Delta\alpha_w = 0$ only at inference. Success remains high at 0.35 and 0.50 m/s, but falls to zero at 0.60 m/s despite substantial row progress, while follow-distance MAE rises to 1.46 m. With the experts and

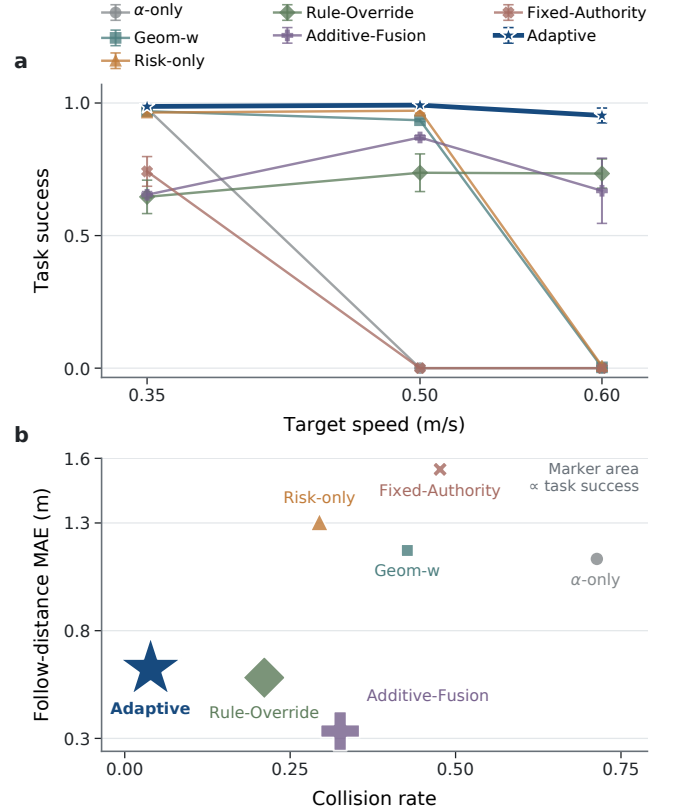


Fig. 3. Main performance on the held-out Stage-4 evaluation. (a) Task success versus target speed. (b) Follow-distance MAE versus collision rate at 0.60 m/s; marker area encodes task success.

policy checkpoint fixed, this comparison supports the learned correction’s role in sustaining target following under the highest tested conflict. Avoid and the gate observe body-frame velocity, while Follow responds to the relative target state; the highest speed therefore stresses the complete system.

C. Structural alternatives. Additive-Fusion maintains low follow-distance error but incurs substantially more collisions, exposing an unresolved safety–following trade-off. Fixed-Authority fails to complete the task at the higher speeds, indicating the limits of one global allocation across conditions. Representative trajectories are shown in Fig. 4(b).

D. External alternatives. We compare against Monolithic PPO as the primary learning baseline and a DWA-style velocity search as a planning baseline; both use the same command interface and target information as Adaptive.

Monolithic PPO. The policy outputs the complete high-level command without Follow/Avoid decomposition. It shares the base observations, spatial map encoder family, command limits, post-processing, and frozen locomotion with Adaptive. Mono and the gate have similar actor–critic parameter counts (1.03M vs. 1.06M), while Adaptive additionally uses the frozen Avoid expert. Mono is trained for up to 61.4M interactions under a curriculum whose progression depends on task performance, with one training seed and three evaluation seeds.

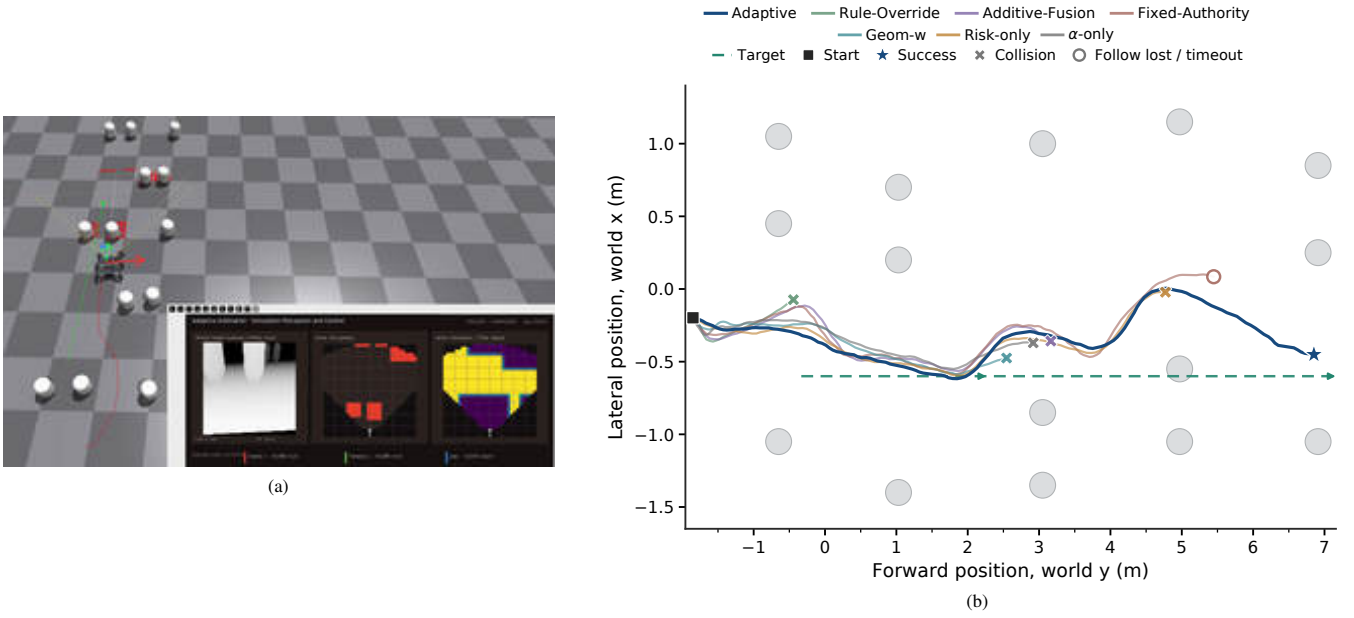


Fig. 4. Training and evaluation scenes. (a) Representative training scene and policy inputs. (b) Representative trajectories at 0.60 m/s; obstacles show the nominal geometry of the held-out Stage-4 layout.

At the matched 24.6M budget, Mono has zero task success at all three speeds, whereas Adaptive exceeds 95% (Table IV). Extended training raises Mono to 94.5% success at 0.50 m/s, confirming that the complete task is learnable without expert decomposition. Performance nevertheless remains uneven in this training run: at 0.35 m/s, within the training range, success falls from 57.3% at 49.2M interactions to zero at 61.4M. At 49.2M and 0.60 m/s, 60.4% of episodes complete all rows without collision, yet task success remains zero and follow-distance MAE reaches 1.25 m. Thus, obstacle traversal can be acquired without consistently satisfying the following objective.

DWA-style velocity search. DWA uses the same map, target, command limits, post-processor, and low-level policy as Adaptive. It searches reachable commands at 10 Hz, scoring

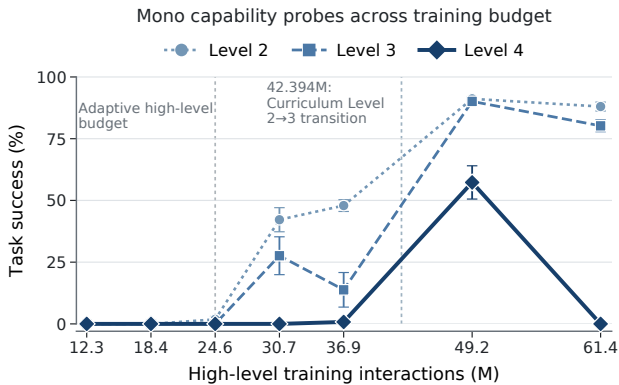


Fig. 5. Mono capability emergence. Probes use frozen checkpoints at curriculum levels 2–4 and 0.35 m/s, with policy updates disabled. Error bars show sample SD across three evaluation seeds for one training seed; the vertical lines mark the Adaptive budget and curriculum transition.

TABLE II
PERFORMANCE ON THE HELD-OUT STAGE-4 EVALUATION. VALUES ARE MEAN $\pm$ SD; SPEED IN M/S. **BOLD MARKS THE BEST MEAN PER SPEED AND METRIC**

Spd.	Method	Succ. [%] $\uparrow$	Row prog. [%] $\uparrow$	Coll. [%] $\downarrow$	MAE [m] $\downarrow$
0.35	α -only	97.9 $\pm$ 0.5	99.6 $\pm$ 0.1	2.1 $\pm$ 0.5	0.217 $\pm$ 0.001
	Geom-w	96.9 $\pm$ 1.4	99.2 $\pm$ 0.6	3.1 $\pm$ 1.4	0.171 $\pm$ 0.001
	Risk-only	96.4 $\pm$ 1.2	99.3 $\pm$ 0.2	3.6 $\pm$ 1.2	0.164 $\pm$ 0.001
	Rule-Override	64.6 $\pm$ 6.3	82.1 $\pm$ 4.0	34.6 $\pm$ 6.6	0.164 $\pm$ 0.008
	Additive-Fusion	65.4 $\pm$ 1.2	84.2 $\pm$ 1.0	34.4 $\pm$ 1.4	0.102 $\pm$ 0.001
	Fixed-Authority	74.2$\pm$5.6	89.9$\pm$3.0	24.5$\pm$6.1	0.390 $\pm$ 0.009
	Adaptive	98.7$\pm$0.9	99.8$\pm$0.1	1.3$\pm$0.9	0.120 $\pm$ 0.000
0.50	α -only	0.0 $\pm$ 0.0	93.0 $\pm$ 1.2	30.7 $\pm$ 6.4	0.969 $\pm$ 0.035
	Geom-w	93.5 $\pm$ 0.5	97.2 $\pm$ 0.7	6.5 $\pm$ 0.5	0.430 $\pm$ 0.001
	Risk-only	97.1 $\pm$ 0.9	99.0 $\pm$ 0.5	2.9 $\pm$ 0.9	0.405 $\pm$ 0.001
	Rule-Override	73.7 $\pm$ 7.1	90.0 $\pm$ 2.8	24.7 $\pm$ 7.3	0.381 $\pm$ 0.013
	Additive-Fusion	87.0 $\pm$ 0.9	94.6 $\pm$ 0.4	12.5 $\pm$ 0.8	0.177 $\pm$ 0.002
	Fixed-Authority	0.0$\pm$0.0	43.6$\pm$0.5	50.8$\pm$4.1	1.220 $\pm$ 0.020
	Adaptive	99.2$\pm$0.8	99.9$\pm$0.2	0.8$\pm$0.8	0.355 $\pm$ 0.003
0.60	α -only	0.0 $\pm$ 0.0	79.4 $\pm$ 0.9	71.4 $\pm$ 3.6	1.133 $\pm$ 0.012
	Geom-w	0.3 $\pm$ 0.5	74.6 $\pm$ 1.8	42.7 $\pm$ 8.4	1.172 $\pm$ 0.015
	Risk-only	0.8 $\pm$ 0.8	71.5 $\pm$ 0.2	29.4 $\pm$ 7.1	1.305 $\pm$ 0.031
	Rule-Override	73.4 $\pm$ 5.5	86.1 $\pm$ 2.7	21.1 $\pm$ 4.8	0.582 $\pm$ 0.019
	Additive-Fusion	66.9 $\pm$ 12.3	87.4 $\pm$ 5.0	32.6 $\pm$ 12.6	0.334 $\pm$ 0.040
	Fixed-Authority	0.0$\pm$0.0	39.9$\pm$0.1	47.7$\pm$2.8	1.549 $\pm$ 0.004
	Adaptive	95.3$\pm$2.8	98.5$\pm$0.8	3.9$\pm$2.7	0.623 $\pm$ 0.004

collision, risk, target distance and bearing, progress, and smoothness. Tuning on validation trials selected a $7 \times 5 \times 5$ grid and 1.2 s horizon. The best tested setting achieves 8.2% success at 0.35 m/s and zero at 0.60 m/s (Table V). Across the tested cost settings, changes in obstacle progress and following error do not yield both low collision and high task success.

E. Real-robot deployment. We deploy the policy on a hexapod in a narrow indoor corridor with two staggered

TABLE III

ABLATION OF THE LEARNED CORRECTION ($\Delta\alpha_w = 0$ AT INFERENCE).
SUCCESS, ROW PROGRESS, AND COLLISION ARE PROPORTIONS; SPEED
IS IN M/S

Spd.	Success $\uparrow$	Row prog. $\uparrow$	Coll. $\downarrow$	MAE [m] $\downarrow$	$\Delta\alpha_w$ (pred.)
0.35	0.974	0.993	0.023	0.176	0.066
0.50	0.982	0.992	0.016	0.438	0.067
0.60	0.000	0.739	0.133	1.463	0.066

obstacle rows. Their passable gaps alternate left and right, so the robot must sidestep through each row while keeping the walking person in view. Hardware rows replace the regular simulated primitives with boxes and cones, testing transfer beyond the simulated obstacle representation. YOLOv8 is used only for target detection and relative goal estimation; obstacle geometry is derived from stereo depth after masking depth points associated with the detected target. The remaining valid depth points are back-projected using camera intrinsics, transformed to robot-aligned coordinates, filtered by height, and rasterized into the occupancy channel

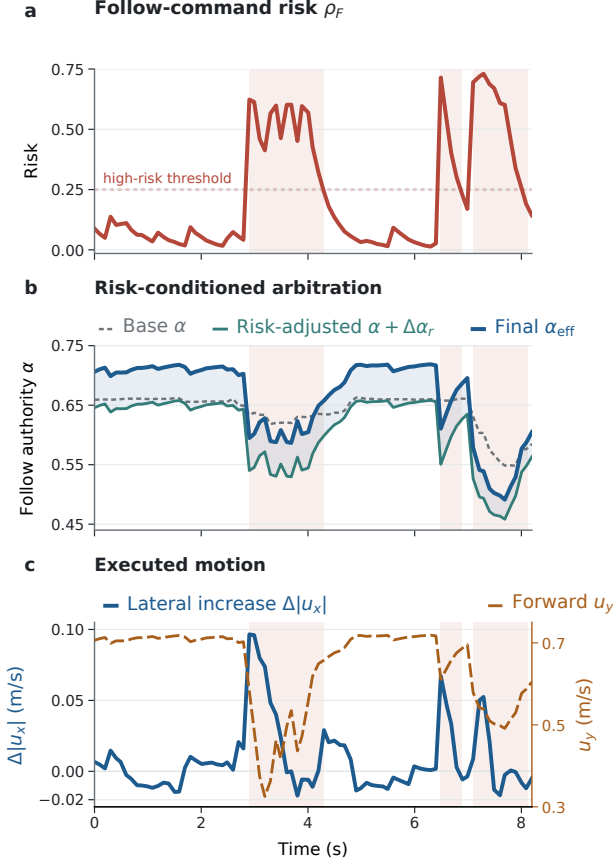


Fig. 6. Representative real-robot arbitration. (a) Risk derived from stereo depth peaks during conflict. (b) The analytic risk term reduces Follow authority; blue shading shows its partial restoration by the learned correction. (c) Forward and lateral motion; $\Delta|u_x|$ is relative to the pre-conflict median.

TABLE IV

HELD-OUT STAGE-4 TASK SUCCESS (%); SPEEDS IN M/S

Policy	Budget (M)	0.35	0.50	0.60
Adaptive	24.576	98.7 $\pm$ 0.9	99.2 $\pm$ 0.8	95.3 $\pm$ 2.8
Mono 2000	24.576	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Mono 4000	49.152	57.3 $\pm$ 6.7	94.5 $\pm$ 0.0	0.0 $\pm$ 0.0
Mono 5000	61.440	0.0 $\pm$ 0.0	93.0 $\pm$ 2.7	1.3 $\pm$ 0.9

Adaptive: 12.288M Avoid + 12.288M gate interactions (shared low-level training excluded); Mono uses one training seed. Entries are mean $\pm$ sample SD over three evaluation seeds; each Mono row uses the same checkpoint across speeds.

TABLE V

DWA-STYLE VELOCITY SEARCH

Search bias	Spd.	Success $\uparrow$	Coll. $\downarrow$	Row prog. $\uparrow$	MAE [m] $\downarrow$
Safety-biased	0.35	0.053	0.612	0.138	0.678
	0.60	0.000	0.838	0.000	1.028
Balanced	0.35	0.026	0.938	0.188	0.375
	0.60	0.000	0.969	0.375	0.592
Tracking-biased	0.35	0.082	0.744	0.344	0.528
	0.60	0.000	0.881	0.500	0.690

of the local map. The local map is updated from current and short-term depth observations in the robot-fixed frame, while m_t separately retains scalar risk history. No persistent global obstacle map is maintained. Simulation uses actor rasterization in place of stereo depth. Adaptive and Rule-Override use identical target detection, local map generation, and perception settings. Under the staggered-row protocol, Adaptive achieves higher task success (Table VI) and lower mean follow-distance MAE than Rule-Override (0.52 versus 1.21 m). Fig. 6 shows a representative arbitration trace; Fig. 7 shows the real-robot and onboard views.

A sharp-turn corridor joins two row segments at 90°. Adaptive succeeds in 18/20 trials without scene-specific switching, bringing its overall success to 37/40 across both scenes (Table VI). The three Adaptive failures comprise two obstacle contacts and one target loss. Either event makes the trial unsuccessful; subsequent recovery does not change this classification. The supplementary video shows the sequence.

VII. DISCUSSION AND LIMITATIONS

Role of adaptive arbitration. Lower collision rates, substantial obstacle progress, and low follow-distance error can each coexist with task failure. With the experts fixed, arbitration variants show different coordination failure patterns. More directly, disabling the learned correction in the same Adaptive checkpoint reduces task success to zero at the highest tested speed. Mono confirms that the complete task is learnable without expert decomposition, but its evaluated run does not consistently maintain obstacle traversal and sustained following together across speeds. These findings support learning how authority should change with context.

Scope and limitations. Lateral-only Avoid assumes a feasible bypass observable in the forward map. The evaluated obstacles are static; interaction with dynamic obstacles is outside the present scope. Solid transverse walls, dead ends, concave traps, or a target stopping before a barrier may

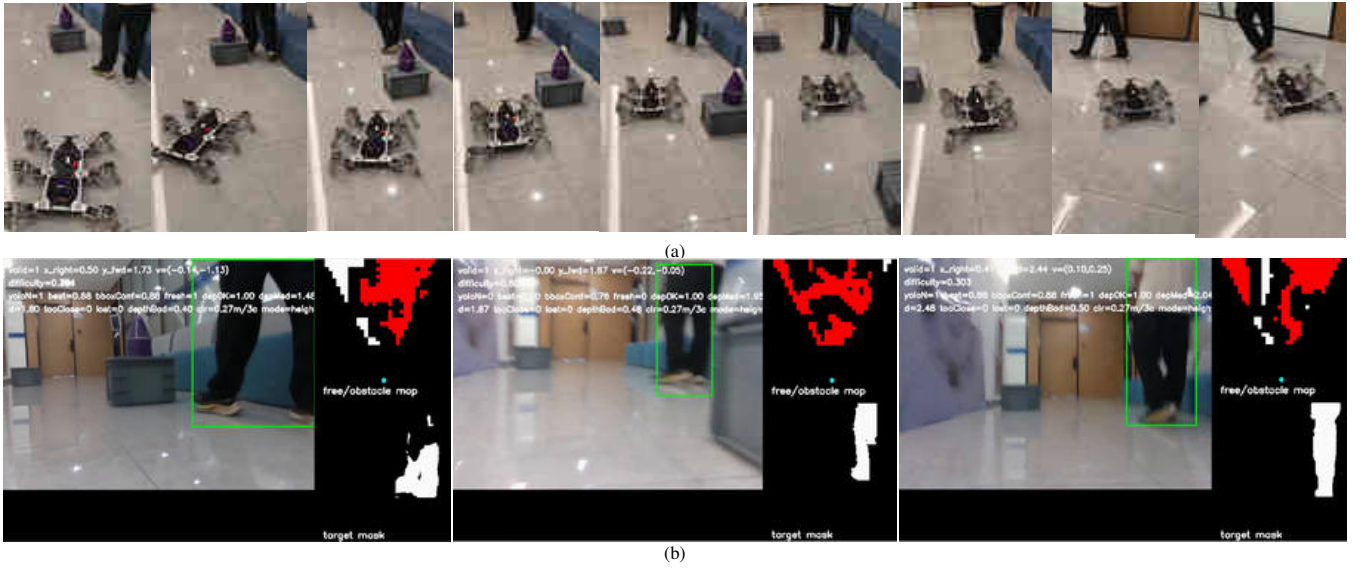


Fig. 7. Real-robot deployment views. (a) The hexapod follows the walking target while sidestepping around staggered corridor obstacles. (b) Three representative onboard times (frames 1, 3, and 5) show the target state and local occupancy-free-space map derived from stereo depth and used by Adaptive.

TABLE VI
REAL-ROBOT RESULTS (60 TRIALS)

Method	Scene	Trials	Success	Coll.	Lost	Rate
Adaptive	Staggered rows	20	19	1	0	95%
Rule-Override	Staggered rows	20	12	6	2	60%
Adaptive	Sharp target turn	20	18	1	1	90%
Adaptive	Overall	40	37	2	1	92.5%

require stopping, backward motion, or higher-level replanning; the evaluated controller has no dedicated dead-end recovery or global replanning mechanism. Directional risk measures proximity rather than swept-footprint clearance, and the bounded convex blend does not certify collision-free execution.

VIII. CONCLUSION

We presented adaptive command-space arbitration for target following under passability mismatch, focusing learning on the coordination of fixed Follow and Avoid proposals. Controlled comparisons show that Additive-Fusion and Fixed-Authority do not maintain comparable task performance across the tested conditions, while extended Monolithic PPO training confirms that the complete task can be learned without expert decomposition. Adaptive achieves 95.3% success on the held-out evaluation at 0.60 m/s and succeeds in 37 of 40 hardware trials across staggered-row and sharp-turn corridors. Together, these results support context-dependent authority allocation for balancing obstacle negotiation with sustained target following in the tested conditions.

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